

A Detailed Descriptions of Annotation Clustering Module (ACM) Components

This appendix provides the full details for Stage 1 (Cluster Number Determination) and Stage 3 (Outlier Removal) of the Annotation Clustering Module, which were briefly introduced in Section 3.1 of the main paper.

A.1 Cluster Number Determination Method

Preparation: the Segment Anything Model (SAM) (?) is an impressive segmentation model renowned for its robust segmentation capabilities, widely applied across various computer vision domains. In our work, as a pre-processing step we also leverage SAM to refine worker annotations. Specifically, for a single image A_i , by feeding every worker ROI r_m and the image itself as prompts into SAM, it utilizes the raw image features to produce a SAM-refined ROI r_m^{sam} and the corresponding confidence conf_m^{sam} , leading to A_i^{sam} .

Algorithm 1: Determine Number of Clusters

Input: A_i : Set of annotations in T_i , τ : Similarity threshold

Output: Number of clusters K

```

1: Initialize connection matrix  $\mathcal{M}$  with zeros
2:  $K \leftarrow 0$ 
3: for each pair  $(a_m, a_n)$  in  $A_i$  do
4:    $\mathcal{M}_{m,n} \leftarrow \text{IoU}(a_m, a_n)$ 
5: end for
6: for each  $a_m$  in  $A_i$  do
7:   if  $a_m$  is not visited then
8:      $U \leftarrow \text{UniqueWorkerDFS}(a_m, \mathcal{M}, \tau)$ 
9:      $K \leftarrow K + 1$ 
10:  end if
11: end for
12: return  $K$ 
```

Algorithm 2: UniqueWorkerDFS

Input: $node$: Start node, $\mathcal{M}$: Affinity matrix, τ : threshold

Output: U : List of visited nodes

```

1: Initialize an empty list  $U$  and a stack  $S$ 
2: Add  $node$  to  $U$  and  $S$  respectively
3: while  $S$  is not empty do
4:   Pop  $currentNode$  from  $S$ 
5:   for each  $neighbor$  of  $currentNode$  in  $\mathcal{M}$  do
6:     if  $\mathcal{M}_{currentNode, neighbor} > \tau$  and  $neighbor \notin U$  then
7:       Add  $neighbor$  to  $U$ , push  $neighbor$  onto  $S$ 
8:     end if
9:   end for
10: end while
11: for each pair  $(a_m, a_n) \in U$  annotated by the same worker do
12:    $s_m, s_n \leftarrow$  the accumulated affinity of rows of  $a_m, a_n$  in  $\mathcal{M}$ 
13:   Pop  $a_{\min(s_m, s_n)}$  from  $U$ 
14: end for
15: return  $U$ 
```

For the cluster number, either an over-fine-grained or a coarse-grained clustering would lead to misinformation for the latter aggregation steps. Algorithm 1 addresses this issue:

- **Lines 1-2: Initialization.** The algorithm starts by initializing a matrix $\mathcal{M}$ with zeros and setting the cluster counter K to 0.
- **Lines 3-5: Construct Affinity Matrix.** For each pair of annotations (a_m, a_n) in the set of worker annotations A_i , the algorithm calculates an affinity score between their ROIs, via their Intersection over Union (IoU). Note that in this section, when we refer to an annotation, we mean its ROI.
- **Lines 6-9: Connected Component Analysis.** The algorithm conducts a connected component analysis on the matrix $\mathcal{M}$. Each group of connected annotations forms a cluster, denoted as U . To do so, treating an ROI as a node and placing edges between two ROIs with affinity scores larger than τ , we then can use the vanilla depth-first search (DFS) to traverse the un-visited ROIs one-by-one. However, it may absorb two ROIs from the same worker in a single cluster. This apparently violates the worker’s intention (a worker would not repeatedly draw ROIs for the same object). We then propose UniqueWorkerDFS in Algorithm 2, which prevents this over-connection. Specifically, Algorithm 2 (Lines 3-7) performs a DFS traversal by popping the current node, examining its neighbors, and adding those with affinity scores above τ to U and S . Then, for each pair of annotations from the same worker in U , it retains only the one with the higher accumulated affinity score (Lines 8-10 of Algorithm 2).
- **Line 9: Returning the Number of Clusters.** Finally, the algorithm returns the total number of clusters identified, providing a measure of the granularity of the clustering.

A.2 Outlier Removal Method

The medical image annotators are also likely to make faults and give wrong ROIs, especially when the image itself is difficult or the annotator gets tired. Such wrong ROIs typically do not fit any cluster and could be detected as outliers.

The previous section produces a set of clusters as $\{F_1, F_2, \dots, F_K\}$, where $F_k = \{a_1, a_2, \dots, a_{|F_k|}\}$, and $\sum_{k=1}^K |F_k| = |A_i|$. To identify outliers, we first define a Relative Intersection Ratio (RIR) score as

$$RIR(r_m, r_n) = \frac{\text{area}(r_m \cap r_n)}{\text{area}(r_m)}. \quad (1)$$

For a specific ROI r_m within a cluster F_k , we determine whether it is an outlier in terms of two affinity scores.

Score 1: Intra-Cluster affinity. We define the affinity score of an annotation r_m within its own cluster as

$$p_1^m = \max_{r_n \in F_k, n \neq m} (1 - \text{GIoU}(r_m, r_n)). \quad (2)$$

where r_n represents other ROIs in F_k .

Score 2: Inter-Cluster affinity. We further compute its affinity to other clusters as:

$$p_2^m = \frac{1}{K-1} \sum_{k'=1, k' \neq k}^K \sum_{a_n \in F_{k'}} \frac{RIR(r_m, r_n)}{|F_{k'}|}. \quad (3)$$

Intuitively, a ROI that excessively matches other clusters while failing to match the ROIs within its own cluster is considered as an outlier. We then model the event that ROI r_m is an outlier with a Bernoulli distribution $Ber(p_1^n \cdot p_2^n)$. Once ROI is identified as an outlier w.r.t. the Bernoulli event, it is then removed.

B Proof of Theoretical Claims

This appendix provides the detailed derivations and proofs for the theoretical claims presented in Section 3.2 of the main paper, specifically for Theorem 1.

B.1 Proof of Theorem 1

Objective Equivalence. To attain the equivalent formulation, we begin with some auxiliary definitions.

Lemma 1 (A property of the Laplacian Matrix (??)). *For the Laplacian matrix $\mathbb{L}$, assuming its dimension is n , and the value at the i -th row and j -th column is $w_{i,j}$, it satisfies the following property:*

For any vector $f \in \mathbb{R}^n$, we have $f^T \mathbb{L} f = \frac{1}{2} \sum_{i,j=1}^n w_{i,j} (f_i - f_j)^2$.

The detailed proof of this lemma can be found in previous studies (??).

We define the Laplacian matrix of an image as $\mathbb{L} = D - S$. In addition, for a certain clustering result, we introduce the indicator vectors $h_k \in \{h_1, h_2, \dots, h_K\}$ for $k = 1, 2, \dots, K$. Each vector h_k is a $|A_i|$ -dimensional vector, where $|A_i|$ is the number of ROIs collected for the image to be clustered, $h_k = [h_{1,k}, h_{2,k}, \dots, h_{|A_i|,k}]$. We define $h_{m,k}$ as follows:

$$h_{m,k} = \begin{cases} 0 & a_m \notin F_k \\ \frac{1}{\sqrt{\text{CrowdWeight}(F_k)}} & a_m \in F_k \end{cases} \quad (4)$$

which indicates the cluster membership of the ROIs. Then we set the matrix $H \in \mathbb{R}^{|A_i| \times K} = \{h_k\}$ as the composition of such vectors.

Based on the above definitions, we have the following equivalence theorem.

Theorem 1: Minimizing the cut objective in Equation (??) is equivalent to the following optimization objective:

$$\begin{aligned} & \arg \min_{F_1, F_2, \dots, F_K} \text{tr}(H^T \mathbb{L} H) \\ & \text{s.t. } H^T M H = I, \quad H \text{ as defined in Eq.(4).} \end{aligned} \quad (5)$$

where $\text{tr}(\cdot)$ denotes the trace of a matrix.

Proof Sketch. According to the definition of H , $\mathbb{L}$ and M , together with a lemma from the existing literature, we deduct $h_k^T \mathbb{L} h_k = \frac{\text{cut}(F_k, \bar{F}_k)}{\text{CrowdWeight}(F_k)}$, and then $H^T M H = I$. On top of these two properties, the objective transformation is conducted. $\square$

Proof. **Proving $h_k^T \mathbb{L} h_k = \frac{\text{cut}(F_k, \bar{F}_k)}{\text{CrowdWeight}(F_k)}$.** From Lemma 1, we have

$$h_k^T \mathbb{L} h_k = \frac{1}{2} \sum_{m=1}^{|A_i|} \sum_{n=1}^{|A_i|} S_{m,n} (h_{m,k} - h_{n,k})^2$$

Therefore,

$$\begin{aligned} h_k^T \mathbb{L} h_k &= \frac{1}{2} \left(\sum_{\substack{a_m \in F_k \\ a_n \notin F_k}} S_{m,n} \left(\frac{1}{\sqrt{\text{CrowdWeight}(F_k)}} - 0 \right)^2 \right. \\ &\quad \left. + \sum_{\substack{a_m \notin F_k \\ a_n \in F_k}} S_{m,n} \left(0 - \frac{1}{\sqrt{\text{CrowdWeight}(F_k)}} \right)^2 \right) \\ &= \frac{1}{2} \left(\sum_{\substack{a_m \in F_k \\ a_n \notin F_k}} S_{m,n} \frac{1}{\text{CrowdWeight}(F_k)} + \sum_{\substack{a_m \notin F_k \\ a_n \in F_k}} S_{m,n} \frac{1}{\text{CrowdWeight}(F_k)} \right) \\ &= \frac{1}{2} \left(\text{cut}(F_k, \bar{F}_k) \frac{1}{\text{CrowdWeight}(F_k)} + \text{cut}(\bar{F}_k, F_k) \frac{1}{\text{CrowdWeight}(F_k)} \right) \\ &= \frac{\text{cut}(F_k, \bar{F}_k)}{\text{CrowdWeight}(F_k)} \end{aligned} \quad (6)$$

Proving $H^T M H = I$. To simplify the notation, let $(H^T M H)_{pq}$ denote the element in the p -th row and q -th column of the matrix $H^T M H$. Clearly, we have:

$$(H^T M H)_{p,q} = \sum_{m=1}^{|A_i|} \sum_{n=1}^{|A_i|} h_{m,p} M_{m,n} h_{n,q}$$

Because M is a diagonal matrix, the product of this formula is not 0 only when $m = n$, so in fact:

$$(H^T M H)_{p,q} = \sum_{m=1}^{|A_i|} h_{m,p} M_{m,m} h_{m,q}$$

Based on the definition of the h vectors, $h_{m,p}$ and $h_{m,q}$ are non-zero only when the ROI belongs to cluster F_p or F_q . Moreover,

$$h_{m,p} = \frac{1}{\sqrt{\text{CrowdWeight}(F_p)}}$$

$$h_{m,q} = \frac{1}{\sqrt{\text{CrowdWeight}(F_q)}}$$

Thus, the above expression can be rewritten as:

$$(H^T M H)_{p,q} = \sum_{a_m \in F_p \cap F_q} \frac{1}{\sqrt{\text{CrowdWeight}(F_p)}} M_{m,m} \frac{1}{\sqrt{\text{CrowdWeight}(F_q)}}$$

Combining this with the definition of M , we obtain:

For $p = q$,

$$(H^T M H)_{p,q} = \sum_{a_m \in F_p \cap F_q} \frac{1}{\sqrt{\text{CrowdWeight}(F_p)}} M_{m,m} \frac{1}{\sqrt{\text{CrowdWeight}(F_q)}}$$

since $\text{CrowdWeight}(F_p) = \sum_{a_m \in F_p} M_{m,m}$, the above sum equals 1. Therefore,

$$(H^T M H)_{p,p} = 1$$

For $p \neq q$,

$$(H^T M H)_{p,q} = 0$$

since no ROI simultaneously belongs to different clusters F_p and F_q . Thus, we have proven that $H^T M H = I$.

Proving the objective equivalence. From Equation (6) and $H^T M H = I$, we can derive

$$\begin{aligned} \text{CrowdCut}(F_1, F_2, \dots, F_K) &= \sum_{k=1}^K h_k^T \mathbb{L} h_k \\ &= \sum_{k=1}^K (H^T \mathbb{L} H)_{k,k} \\ &= \text{tr}(H^T \mathbb{L} H) \end{aligned} \quad (7)$$

where $\text{tr}(\cdot)$ denotes the trace of a matrix. And H satisfies $H^T M H = I$.

Therefore, our optimization objective can be formulated as:

$$\begin{aligned} \arg \min_{F_1, F_2, \dots, F_K} \quad & \text{tr}(H^T \mathbb{L} H) \\ \text{s.t.} \quad & H^T M H = I, \quad H \text{ as defined in Eq.(4)} \end{aligned} \quad (8)$$

□

Objective Relaxation. The formulation attained in last subsection is still NP-hard. To find a heuristic for an efficient solution, we adopt a general relaxation method (?), which is to abandon the discreteness condition of h_i 's (its binary values by Eq. (4)). Instead we allow h_i to take any value in $\mathbb{R}^{|A_i|}$. This leads to the relaxed optimization problem.

$$\begin{aligned} \arg \min_{H \in \mathbb{R}^{|A_i| \times K}} \quad & \text{tr}(H^T \mathbb{L} H) \\ \text{s.t.} \quad & H^T M H = I. \end{aligned} \quad (9)$$

We can further reparameterize this optimization problem into a standard matrix eigenvalue decomposition problem. Specifically, let $H = M^{-\frac{1}{2}} R$, then:

$$\begin{aligned} H^T \mathbb{L} H &= R^T M^{-\frac{1}{2}} \mathbb{L} M^{-\frac{1}{2}} R, \\ H^T M H &= R^T R = I. \end{aligned}$$

Thus, the optimization objective becomes:

$$\begin{aligned} \arg \min_{R \in \mathbb{R}^{|A_i| \times K}} \quad & \text{tr}(R^T M^{-\frac{1}{2}} \mathbb{L} M^{-\frac{1}{2}} R) \\ \text{s.t.} \quad & R^T R = I \end{aligned} \quad (10)$$

Note that this is a standard trace minimization problem. According to the Rayleigh entropy theorem (?), the solution R of the above objective is the eigenvectors corresponding to the first K smallest eigenvalues of the matrix $M^{-\frac{1}{2}} \mathbb{L} M^{-\frac{1}{2}}$ by columns. After obtaining R through eigenvectors, H can be obtained as $H = M^{-\frac{1}{2}} R$.

Now the solution H denotes the strength of a_m 's over the K clusters. Since H is no longer binary, to attain the clustering membership, a naive way is to set the cluster of a_m as $\max_k h_{m,k}$. However, this ignores the strength similarity of a_m 's. Thus, we choose the commonly used method in spectral clustering (?) to decide the clusters. The matrix H with dimension $|A_i| \times K$ is treated as the feature matrix of a_m 's $\in A_i$, and we perform K-means clustering on the rows of matrix H , which represent the a_m elements, according to these features.

C Parameter Sensitivity Analysis

This appendix details the parameter sensitivity experiment for the STARLET framework, briefly mentioned in Section 5.4 (Ablation Study) of the main paper.

The STARLET framework introduces one main parameter: the similarity threshold τ used in Algorithm 1 to determine the initial number of clusters K . By default, τ is set to 0.2. To evaluate its impact, we report STARLET's performance across a range of τ values, measuring both ROI quality (Braylan's F1-score and One2one F1-score) and label quality (Precision, Recall, F1-score). The results are visualized in Figure 1.

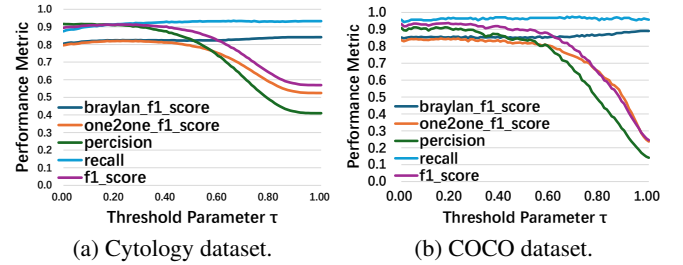


Figure 1: The effect and sensitivity of τ on performance metrics for the Cytology dataset (left) and COCO dataset (right).

As shown in Figure 1:

- When τ increases, the precision and One2one F1-score tend to slowly decrease, while the F1-score for label aggregation also shows a minor decrease.
- Conversely, Braylan's F1-score for ROI quality remains largely stable or even slightly increases, and recall also shows a slight increase.

This behavior can be attributed to the role of τ in determining the number of clusters. A larger τ implies a more stringent similarity requirement for annotations to be considered connected within a component. This often leads to a larger number of smaller, more fine-grained clusters (K increases). A higher K means more predicted objects, which can increase the chance of matching ground truth objects (thus improving recall) but may also lead to more false positives or over-segmentation, thereby reducing precision and One2one F1-score. Braylan's F1-score, which considers the overall overlap regardless of one-to-one mapping, is less sensitive to such over-segmentation. This observation also implicitly validates the necessity of the One2one F1-score as a distinct metric for a more precise evaluation of ROI quality, especially in scenarios with varying cluster granularity.

Based on these results, we chose $\tau = 0.2$ as the default value, as it provides a satisfying balance across all evaluated metrics.